# Attention-Enhanced DCNN-BiLSTM-CTC with Data Augmentation: An End-to-End Architecture for Robust CAPTCHA Recognition

Chen Tianhao (A0314509H), Ding Zhe (A0287899R), Huynh Vu Hoang (A0317306L)
Kin Wei Qi Vanessa (A0258189L), Wu Jiawei (A0281449W)

## Abstract

The challenge of robust, high-accuracy, end-to-end CAPTCHA image detection necessitates advanced sequence recognition methods. This study addresses this need by utilizing a foundational DCNN-BiLSTM-CTC architecture, which we significantly enhanced.

Our key architectural modifications included the integration of Residual layers and Multi-Head Attention to improve convolutional feature extraction and sequential context modeling. To directly combat model overfitting, we implemented data augmentation, creating diverse training samples in every epoch. Further performance gains were achieved through meticulous hyperparameter tuning (e.g., beam width optimization, model size reduction) and the application of the AdamW optimizer with AMSGrad.

The final model attained an overall 67.90%-CAPTCHA accuracy and 92.07%-character accuracy. Analysis revealed common confusions between visually or typographically similar characters, namely (0, o), (1, i, l) and (5, s).

## Pre-Processing

The initial dataset for this project comprised 10,000 CAPTCHA images, partitioned into 8,000 samples for training and 2,000 for testing.

To ensure compatibility with the model architecture, all images were first standardized. This involved resizing every image to a constant dimension and converting the input from RGB to grayscale mode.

The grayscale image data was converted into PyTorch tensors, followed by a critical normalization step. Pixel values, initially ranged from [0, 1], were linearly scaled to the symmetric interval [-1, 1]. This normalization is essential for stabilizing and accelerating neural network training.

The ground-truth labels were extracted directly from the PNG filenames. We also implemented a case-insensitive mapping, where both uppercase ('A') and lowercase ('a') characters were converted to the same class identifier. Finally, ten duplicate images, identified by the (1) suffix in their filenames, were removed from the dataset to guarantee a non-redundant training set.

## Minimally-Viable Products (MVP)

Our team created two MVPs: an RCNN-LSTM-CTC model and a basic DCNN baseline. The DCNN was a purely convolutional network focused on deep feature extraction from the image. The RCNN-LSTM-CTC, in contrast, combined CNN features with an LSTM layer to explicitly model the sequential nature of CAPTCHA characters. Both architectures employed the Connectionist Temporal Classification (CTC) loss to allow end-to-end training without needing character segmentation. While both showed promise in character recognition, they exposed the critical challenge: a single character prediction error results in a sequence failure. **Both models achieved a CAPTCHA accuracy of about 40-50% and character accuracy of about 80%.** We chose to improve on the DCNN baseline as it performed better than the RCNN baseline.

## Methods of Improvements

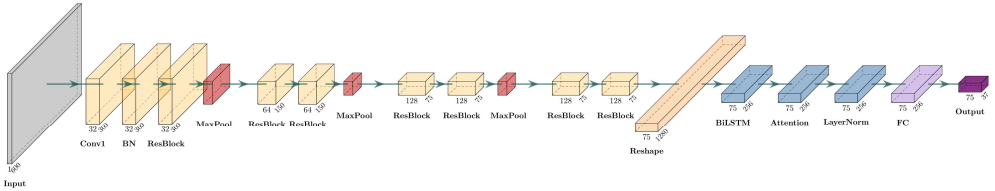
From the basic DCNN model, we implemented five main changes, which significantly improved performance. The changes were: Residual layers, Bidirectional LSTM, Multi-Head Attention, Data Augmentation and AdamW Optimizer with AMSGrad & Learning Rate Scheduling.

**Residual Layers**
Integrating Residual Blocks into the CNN feature extractor allows the model to be built much deeper (4 blocks) without the training issues of vanishing gradients, ensuring that crucial low-level features are preserved and propagate easily through the network.

**Table 1.** Final Model Results

| Set | Level | Accuracy | Precision | Recall |
|---|---|---|---|---|
| Training | Character | 96.75% | 99.62% | 97.11% |
| | CAPTCHA | 84.52% | - | - |
| Testing | Character | 92.07% | 99.46% | 92.57% |
| | CAPTCHA | 67.90% | - | - |

**Figure 1.** Visualization of the Final Model Architecture




## Methods of Improvements

**Bidirectional LSTM (BiLSTM)**
Replacing the DCNN's final classification layer with a BiLSTM explicitly solves the sequential dependency problem in CAPTCHA recognition. By reading the feature sequence both left-to-right (forward) and right-to-left (backward), it captures the complete context of a character's neighbors, which is vital for accurate decoding in messy or connected CAPTCHAs.

**Multi-Head Attention**
Adding a Multi-Head Self-Attention layer after the BiLSTM allows the model to dynamically weigh the importance of every feature across the entire sequence when predicting each character. It effectively filters noise from the background and captures long-range dependencies (e.g., relating the first character to the last) far better than the LSTM alone.

**On-the-Fly Data Augmentation**
Implementing the data augmentation strategy to generate "new" training samples every epoch directly tackles the problem of overfitting. By forcing the model to learn features across a much wider and more varied distribution of images, it dramatically improves the generalization and robustness of the model to unseen CAPTCHA styles.

**AdamW Optimizer with AMSGrad & Learning Rate Scheduling**
AdamW improves regularization over standard Adam, AMSGrad ensures convergence stability and the ReduceLROnPleateau scheduler guarantees that the model takes small, precise steps to find the final, best accuracy point without oscillating.

Hyperparameter tuning, like beam width optimization and model size reduction, was also done.

## Results

Overall, **our final model showed significant improvement** from our two MVPs. We went from barely a 50%-CAPTCHA accuracy to 67.90%, and an 80%-character accuracy to 92.07%.

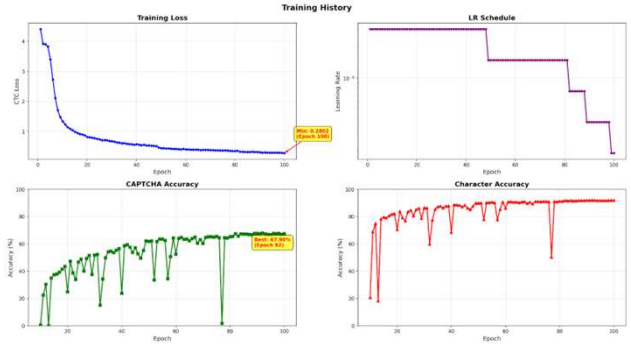
The Test Character Precision is extremely high (99.46%). This means that when the model predicts a character, it is rarely wrong (i.e., it has very few False Positives or "insertions").

The Test Character Recall is much lower (92.57%). This indicates that the model's primary failure mode is missing characters it should have found (i.e., it has many False Negatives or "deletions").

As a result, most sequence-level failures of our model are likely due to the model missing a single character (a deletion/FN), which causes the entire sequence to be marked as incorrect, even if every other character it predicted was right.

Also, there is a large performance gap between the training and test sets. The model achieves 84.52% sequence accuracy on the training data it has already seen, but this drops to 67.90% on new, unseen test data. This means our model may still be overfitting to the training data.

**Figure 2.** Training History of the Final Model




## Discussion

In this study, we decided not to separate the characters in the images and instead process the whole images in the model. This is because newer CAPTCHAs use "anti-segmentation" techniques, such as character adhesion (sticking characters together) , distortion, and "Crowded Characters Together" (CCT). These features make it "harder and harder" to obtain individual characters. Furthermore, traditional segmentation methods had little success when dealing with "adherent characters". Their effect was "not obvious for the seriously adherent and distorted characters", leading to "generally low" success rates.[1]

## Individual Contributions

| Member | Contributions |
|---|---|
| Tianhao | Architected the DCNN-BiLSTM-CTC model. Primary driver for optimizing model performance to the final version. |
| Ding Zhe | Executed comprehensive final performance analysis. Primary driver for the data augmentation in the final version. |
| Vu Hoang | Developed the DCNN MVP. Key contributor to the Pre-processing pipeline setup and data loading procedures. |
| Vanessa | Contributed to the model optimization and performance enhancement strategies. Managed the project timeline, designed the overall poster layout and template, synthesized all research content, and compiled the final poster. |
| Jiawei | Developed the RCNN MVP. Core contributor to the pre-processing pipeline, ensuring data integrity and standardization. |

## References

1. J. Chen, X. Luo, Y. Guo, Y. Zhang, and D. Gong, "A Survey on Breaking Technique of Text-Based CAPTCHA," Security and Communication Networks, vol. 2017, Art. ID 6898617, 15 pages, 2017. DOI: 10.1155/2017/6898617.

© Poster Template by Genigraphics® 1.800.790.4001 www.genigraphics.com